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Then, the magnitude and direction of $\vec{v}_{AB}$ is given by $v_{AB} = \sqrt{v_A^2 + v_B^2 - 2\, v_A v_B \cos\theta}$

and $\tan\beta = \dfrac{v_B \sin\theta}{v_A - v_B \cos\theta}$ (Here β is angle between $\vec{v}_{AB}$ and $\vec{v}_B$)

(i) When $\theta = 0°$, the bodies move along parallel straight lines in the same direction,

We have $v_{AB} = (v_A - v_B)$ in the direction of $\vec{v}_A$. Obviously $v_{BA} = (v_B + v_A)$ in the direction of $\vec{v}_B$.

(ii) When $\theta = 180°$, the bodies move along parallel straight lines in opposite directions,
We have $v_{AB} = (v_A + v_B)$ in the direction of $\vec{v}_A$.

Similarly, $v_{BA} = (v_B - v_A)$ in the direction of $\vec{v}_B$.

(iii) If the two bodies are moving at right angles to each other, then $\theta = 90°$. The magnitude of the relative velocity of A with respect to B $= v_{AB} = \sqrt{v_A^2 + v_B^2}$.

(iv) Consider a person moving horizontally with velocity $\vec{V}_M$. Let rain fall vertically with velocity $\vec{V}_R$. An umbrella is held to avoid the rain. Then the relative velocity of the rain with respect to the person is, **(Figure 2.36)**

$$\vec{V}_{RM} = \vec{V}_R - \vec{V}_M$$

which has magnitude

$$V_{RM} = \sqrt{V_R^2 + V_M^2}$$

and direction $\theta = \tan^{-1}\left(\dfrac{V_M}{V_R}\right)$ with the vertical as shown in Figure. 2.36

In order to save himself from the rain, he should hold an umbrella at an angle θ with the vertical.

EXAMPLE 2.27

Suppose two trains A and B are moving with uniform velocities along parallel tracks but in opposite directions. Let the velocity of train A be 40 km h^{-1} due east and that of train B be 40 km h^{-1} due west. Calculate the relative velocities of the trains

Solution

Relative velocity of A with respect to B, $v_{AB} = 80$ km h^{-1} due east

Thus to a passenger in train B, the train A will appear to move east with a velocity of 80 km h^{-1}

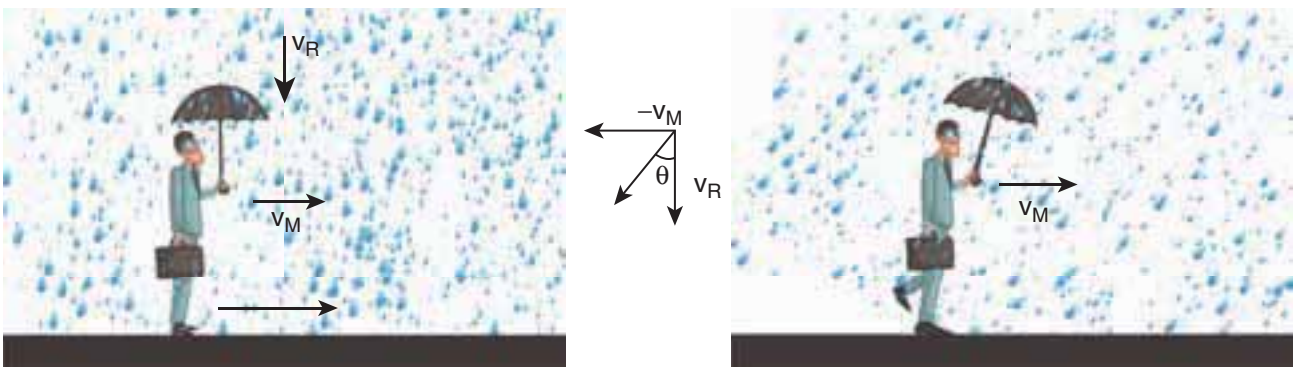


Figure 2.36 Angle of umbrella with respect to rain